Angles and Non-Right Triangles

Concept

Use the unit circle for general angles, and sine/cosine laws for non-right triangles.

Unit circle

$(\cos\theta, \sin\theta)$; $\tan\theta = \sin\theta/\cos\theta$ when defined.

Special angles

Use 30-60-90 and 45-45-90 triangles; apply quadrant signs.

Triangle laws

$a/\sin A = b/\sin B$; $c^2 = a^2 + b^2 - 2ab\cos C$.

SSA ambiguity

Check both B and $180° - B$; keep only valid triangles.

Example 1

Find $\sin150°$, $\cos150°$ and $\tan150°$.

Solution

1. Reference angle $= 30°$; quadrant II has positive sine, negative cosine.

2. $\sin = 1/2$; $\cos = -\sqrt{3}/2$; $\tan = -1/\sqrt{3}$.

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Example 2

Sides 5 and 7 enclose 60°. Find the opposite side c.

Solution

1. $c^2 = 5^2 + 7^2 - 2(5)(7)\cos 60° = 39$.

2. $c = \sqrt{39} \approx 6.24$.

Example 3

Given $A = 30°$, $a = 7$ and $b = 10$, how many triangles exist?

Solution

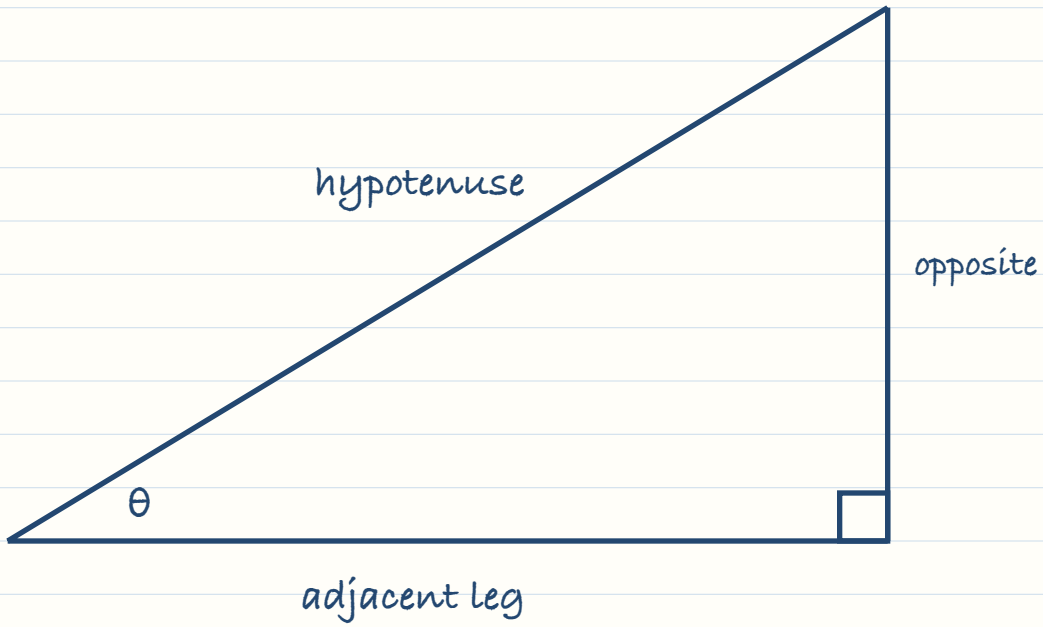
1. $\sin B = 10 \sin 30°/7 = 5/7$.

2. $B \approx 45.58°$ or $134.42°$. Both leave $A + B < 180°$.

3. Two triangles are possible.

Angles and Non-Right Triangles

Visual notes | label before calculating



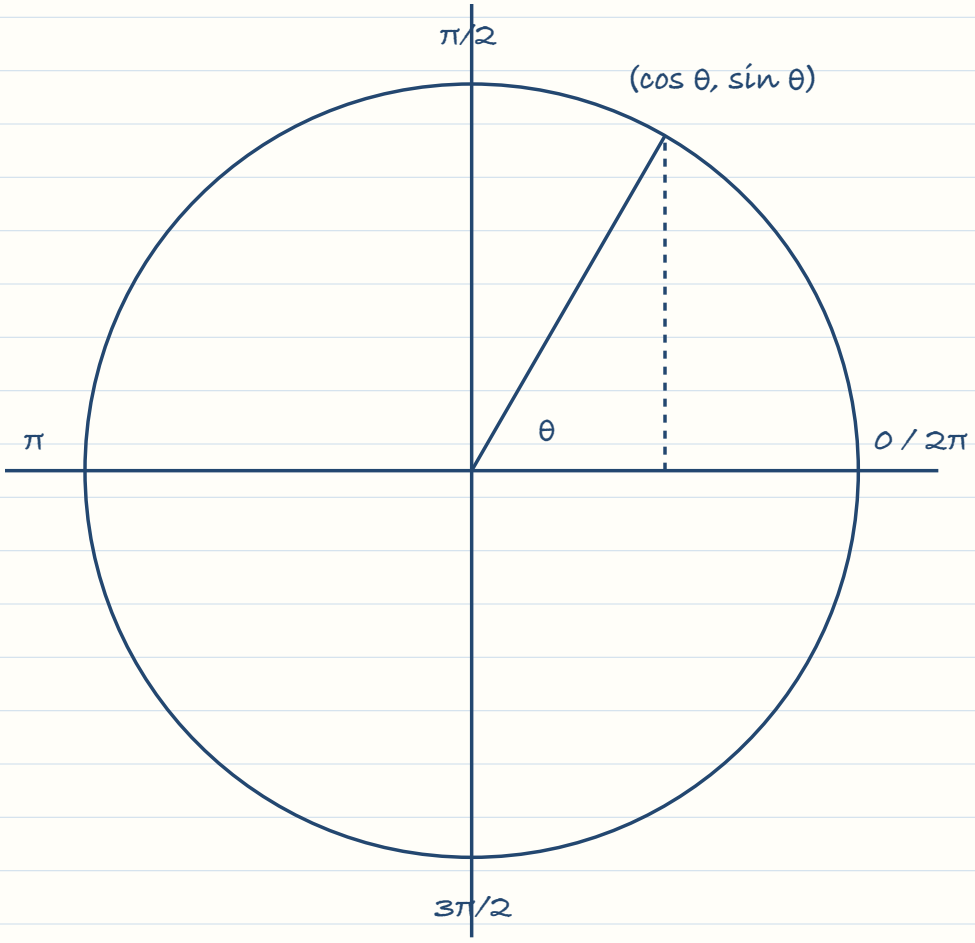
The hypotenuse is opposite the right angle. Opposite and adjacent refer to the selected acute angle θ. A different selected angle changes those two labels.

Pythagoras: $a^2 + b^2 = c^2$. For right triangles, sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, and tan θ = opposite/adjacent.

For a non-right triangle, instead match opposite pairs a/A, b/B and c/C. Sine law needs an opposite pair; cosine law uses an included angle or three sides. Do not apply SOH CAH TOA directly to an arbitrary non-right triangle.

Angles and Non-Right Triangles

Visual notes | unit circle



The radius is 1. The horizontal coordinate is cosine and the vertical coordinate is sine. The drawn angle is π/3 (60°), giving the exact point (1/2, √3/2).

Signs follow the coordinates: upper half means positive sine; right half means positive cosine. Tangent is sin θ/cos θ and is undefined on the vertical axis.

Angles and Non-Right Triangles

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find $\sin 225^\circ$ and $\cos 225^\circ$.

2. Solve $\cos\theta = -1/2$ on $0^\circ \le \theta < 360^\circ$.

3. Find c if $a = 8$, $b = 6$ and included $C = 90^\circ$.

4. If $A = 30^\circ$, $a = 4$ and $b = 10$, how many triangles exist?

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Practice answers | check after trying

1. Both are $-\sqrt{2}/2$, from quadrant III with reference 45°.

2. 120° and 240°.

3. $c = \sqrt{(64 + 36)} = 10$.

4. $\sin B = 10 \sin 30°/4 = 1.25$, impossible, so no triangle.